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Configuring SSH Remote Access on Ubuntu Server

Objective

Set up an Ubuntu Server with SSH (Secure Shell) remote access and connect to it from a host desktop machine using Bridged Network mode.

Operating System Installation

Install Ubuntu Server

1. Start the VM and boot from the Ubuntu Server ISO
2. Follow the installation wizard:
 - Profile setup: Create a user account (e.g., server / server123)
 - Install OpenSSH server: **Check this option** (Important!)
 - Selected software: **None** (Standard Ubuntu Server)
3. Wait for installation to complete and reboot

Network Configuration

Configure Network Mode

For VirtualBox:

1. Power off the VM (if running)
2. Go to **Settings** → **Network**
3. **Adapter 1**: Enable Network Adapter
4. **Attached to**: Select **Bridged Adapter**
5. **Name**: Select your host's physical network adapter (Wi-Fi/Ethernet)
6. Click **OK**

For VMware:

1. Power off the VM
2. Go to **VM Settings** → **Network Adapter**

3. Select **Bridged** mode

4. Click **OK**

Install/Verify SSH Server

Server: Install SSH on the server (if not already installed):

- `sudo systemctl status ssh` – for checking the SSH status.
- `sudo apt update` – updating the your server.
- `sudo apt install openssh-server -y` –installing the SSH.
- `sudo systemctl enable ssh` – enabling the SSH to your unit. (SERVER/DESKTOP)
- `sudo systemctl start ssh` – starting the SSH to your unit. (SERVER/DESKTOP)

Bridged makes the server behave like a separate PC in your LAN.

1. Find the Server's IP

Inside the VM, run:

```
Sample@UBUNTU-DESKTOP:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:9b:8c:ca brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.4/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
        valid_lft 83602sec preferred_lft 83602sec
    inet6 2001:4451:b00:2000:a00:d76c:1af2:a10c/64 scope global temporary dynamic
        valid_lft 179068sec preferred_lft 83386sec
    inet6 2001:4451:b00:2000:a00:27ff:fe9b:8cca/64 scope global dynamic mngtproto kernel_rdr proto kernel_ll
        valid_lft 179068sec preferred_lft 92668sec
    inet6 fe80::a00:27ff:fe9b:8cca/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
Sample@UBUNTU-DESKTOP:~$
```

From the desktop, connect to the server: `ssh username@192.168.1.120`

2. Connect from Desktop

From your host desktop:

```
Sample@UBUNTU-DESKTOP:~$ ssh Server@192.168.1.14
Server@192.168.1.14's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-79-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Sep  8 11:24:38 PM UTC 2025

System load:          0.0
Usage of /:           5.1% of 48.91GB
Memory usage:         5%
Swap usage:           0%
Processes:            139
Users logged in:      1
IPv4 address for enp0s3: 192.168.1.14
IPv6 address for enp0s3: 2001:4451:b00:2000:a00:27ff:fe57:bc4a

Expanded Security Maintenance for Applications is not enabled.

136 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

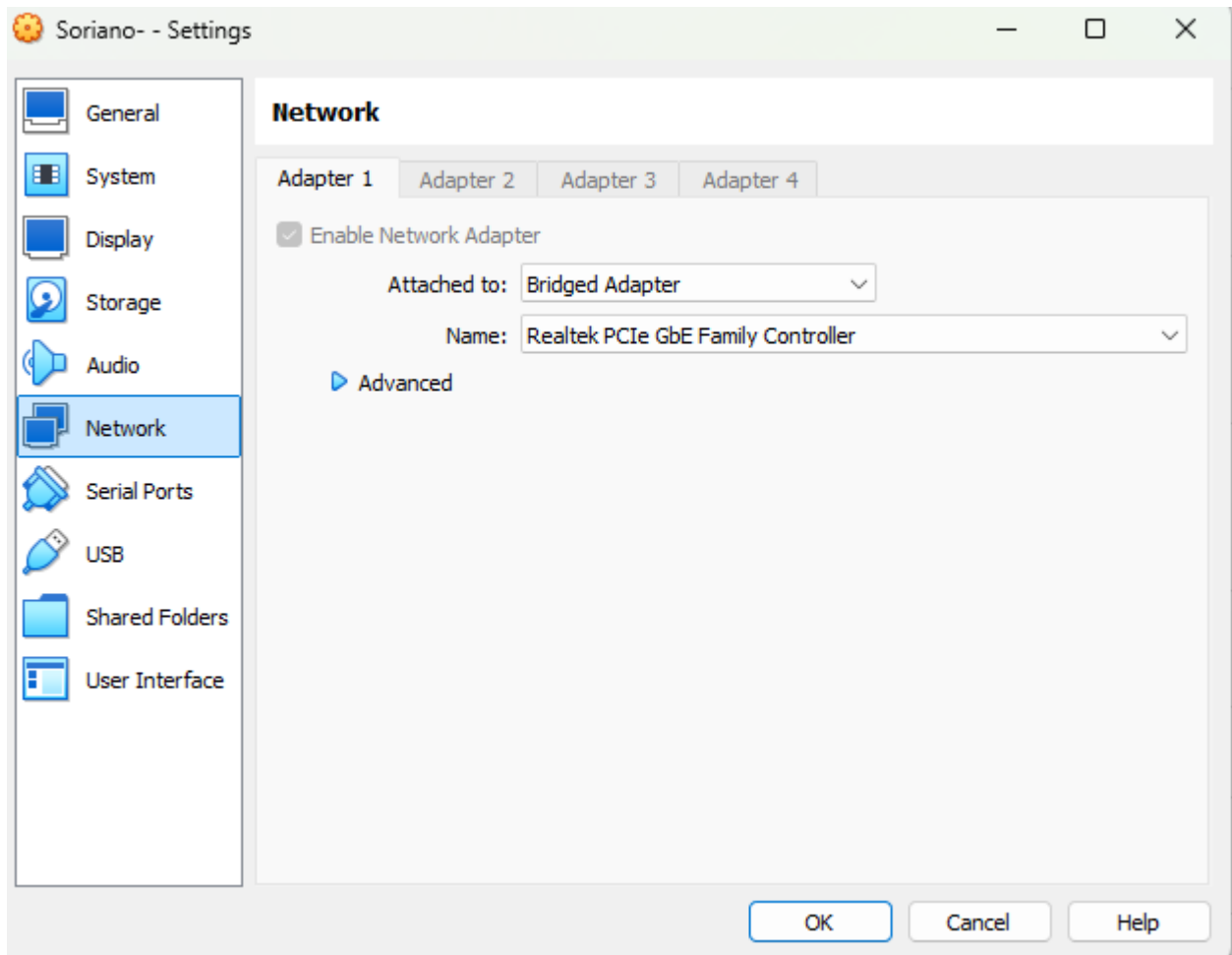
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Sep  8 22:52:07 2025 from 192.168.1.4
Server@UBUNTU-SERVER:~$
```

Submission Requirements

1. Screenshots (labeled clearly):

- VM network settings showing Bridged Adapter



- Server IP address (output of `ip a` or `hostname -I`)

```

Soriano - [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Expanded Security Maintenance for Applications is not enabled.
2 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

richard@ricci:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:80:c5:71 brd ff:ff:ff:ff:ff:ff
    altname enx08002780c571
    inet 172.16.0.174/16 brd 172.16.255.255 scope global dynamic noprefixroute enp0s3
        valid_lft 42sec preferred_lft 34sec
    inet6 fe80::a00:27ff:fe80:c571/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
richard@ricci:~$ sudo systemctl status ssh
[sudo: authenticate] Password:
• ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-09-03 04:10:43 UTC; 38min ago
   Invocation: 3ea90319d7f64cde92e830a6fc24ba7b
   TriggeredBy: • ssh.socket
   Docs: man:sshd(8)
        man:sshd_config(5)
   Process: 1096 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 1132 (sshd)
   Tasks: 1 (limit: 3876)
   Memory: 6.6M (peak: 8.8M)
   CPU: 71ms
   CGroup: /system.slice/ssh.service
           └─1132 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

```

- SSH service status (`sudo systemctl status ssh`)

See <https://ubuntu.com/esm> or run: `sudo pro status`

```
richard@ricci:~$ sudo systemctl status ssh
[sudo: authenticate] Password:
```

```
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
```

```
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
```

```
sudo: maximum 3 incorrect authentication attempts
richard@ricci:~$
```

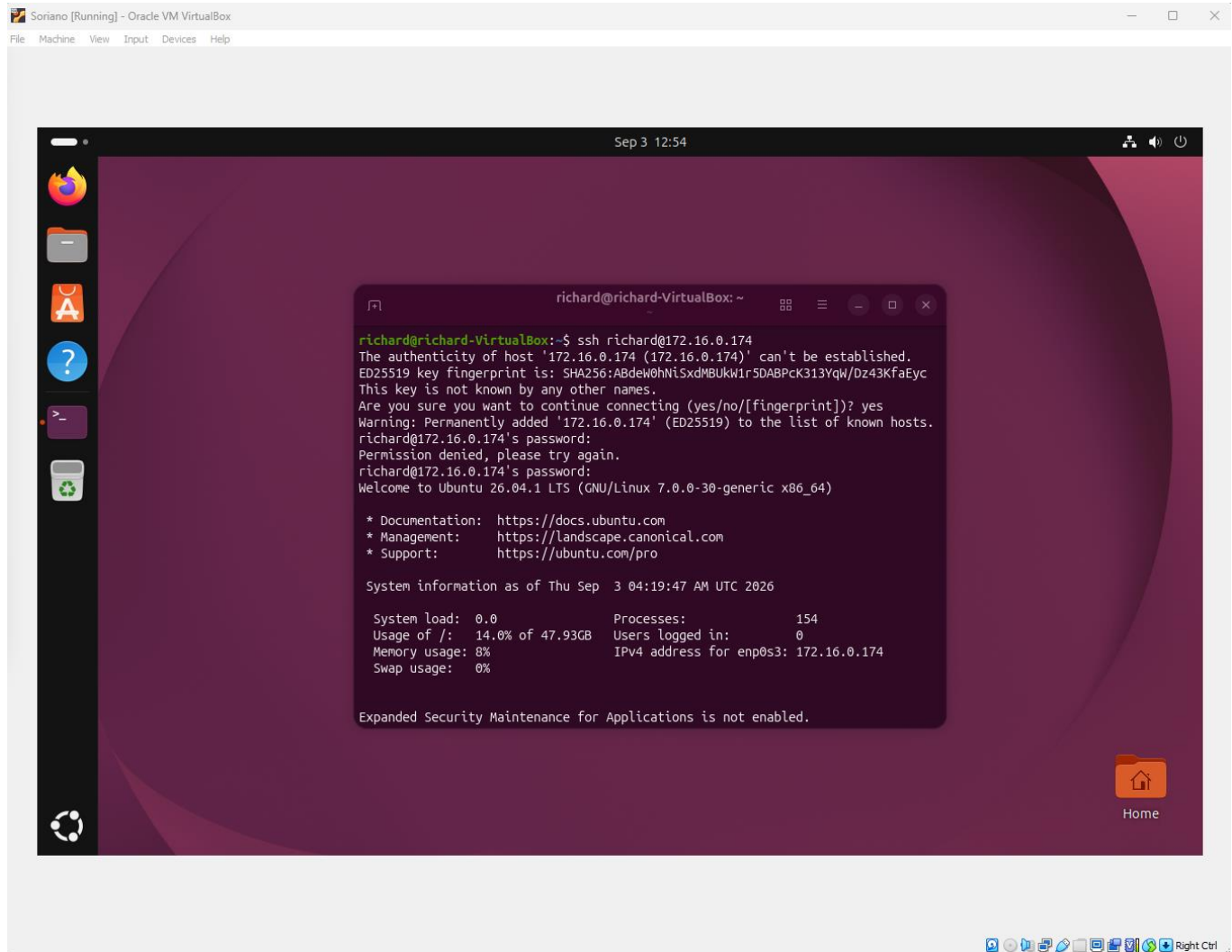
```
richard@ricci:~$ sudo systemctl status ssh
[sudo: authenticate] Password:
```

```
* ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: inactive (dead)
TriggeredBy: ● ssh.socket
            Docs: man:sshd(8)
                  man:sshd_config(5)
```

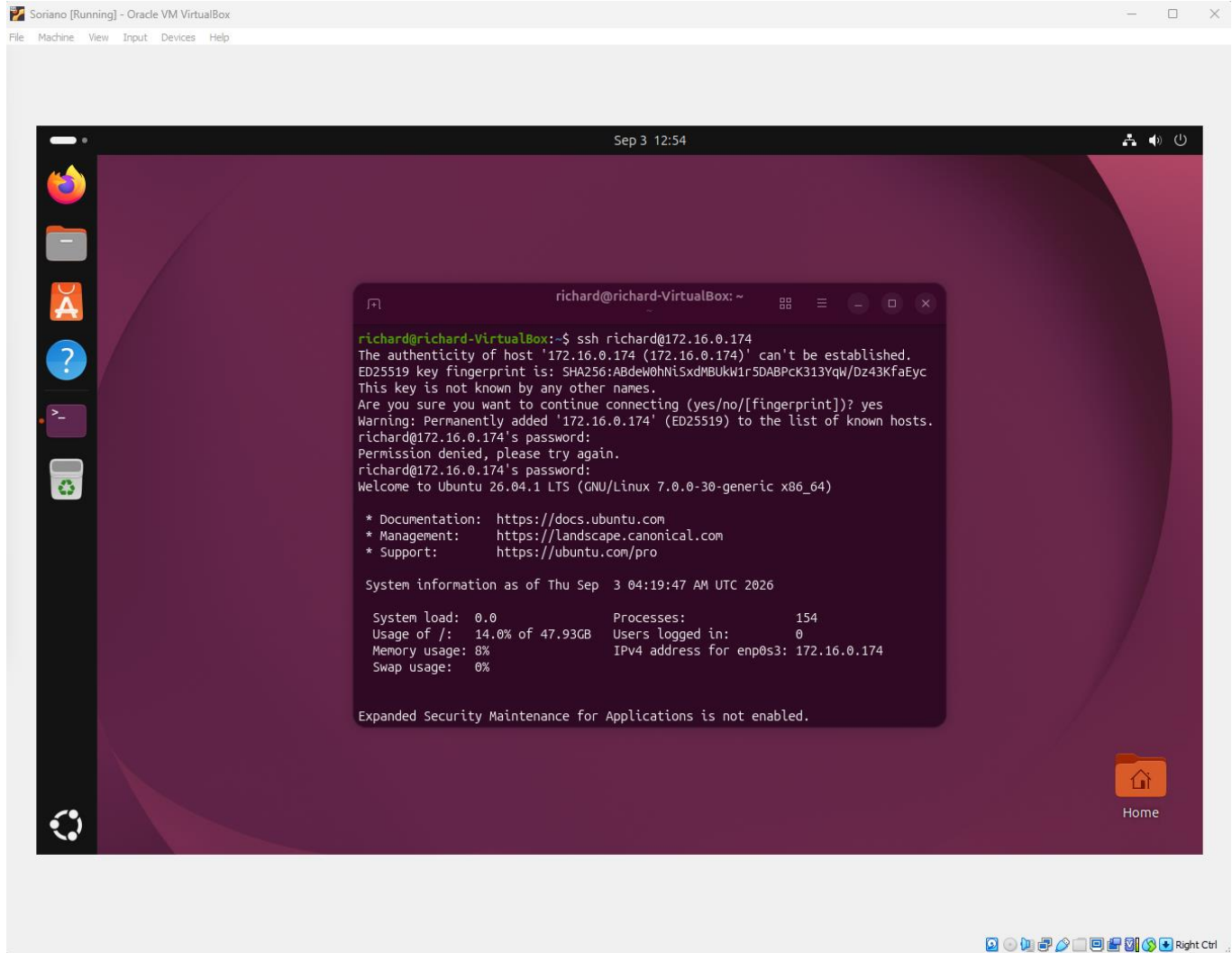
```
richard@ricci:~$ sudo apt update
```

```
Ign:1 http://ph.archive.ubuntu.com/ubuntu resolve InRelease
Ign:2 http://security.ubuntu.com/ubuntu resolve-security InRelease
Ign:3 http://ph.archive.ubuntu.com/ubuntu resolve-updates InRelease
Ign:4 http://ph.archive.ubuntu.com/ubuntu resolve-backports InRelease
Ign:2 http://security.ubuntu.com/ubuntu resolve-security InRelease
Ign:4 http://ph.archive.ubuntu.com/ubuntu resolve-backports InRelease
Ign:3 http://ph.archive.ubuntu.com/ubuntu resolve-updates InRelease
Ign:1 http://ph.archive.ubuntu.com/ubuntu resolve InRelease
Ign:1 http://ph.archive.ubuntu.com/ubuntu resolve InRelease
Ign:2 http://security.ubuntu.com/ubuntu resolve-security InRelease
Ign:3 http://ph.archive.ubuntu.com/ubuntu resolve-updates InRelease
Ign:4 http://ph.archive.ubuntu.com/ubuntu resolve-backports InRelease
Err:4 http://ph.archive.ubuntu.com/ubuntu resolve InRelease
      Temporary failure resolving 'ph.archive.ubuntu.com'
Err:2 http://security.ubuntu.com/ubuntu resolve-security InRelease
      Temporary failure resolving 'security.ubuntu.com'
Err:3 http://ph.archive.ubuntu.com/ubuntu resolve-updates InRelease
      Temporary failure resolving 'ph.archive.ubuntu.com'
Err:1 http://ph.archive.ubuntu.com/ubuntu resolve InRelease
      Temporary failure resolving 'ph.archive.ubuntu.com'
5 packages can be upgraded. Run 'apt list --upgradable' to see them.
Warning: Failed to fetch http://ph.archive.ubuntu.com/ubuntu/dists/resolve/InRelease Tempora
Warning: Failed to fetch http://ph.archive.ubuntu.com/ubuntu/dists/resolve-backports/InReleas
Warning: Failed to fetch http://ph.archive.ubuntu.com/ubuntu/dists/resolve-security/InReleas
Warning: Some index files failed to download. They have been ignored, or old ones used instead.
richard@ricci:~$
```

- Successful SSH connection from host



- Verification commands executed inside VM



2. Submission Format:

- Single PDF file
- Include name, student ID, and date
- All screenshots with brief explanation
- Upload to your portfolio website.

UBUNTU SERVER SSH REMOTE ACCESS LAB

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Course/Year/Section: BSCS3A

Date Submitted: September 9, 2026

Instructor: Johneros P. Puyo

1 OBJECTIVES (5 POINTS)

Check all that apply:

- ✓ Create and configure a Virtual Machine with Ubuntu Server
- ✓ Configure Bridged Network Adapter for network access
- ✓ Determine and verify the server's IP address
- ✓ Install OpenSSH Server and enable the service
- ✓ Establish successful SSH connection from host machine
- ✓ Execute verification commands to confirm remote access

2 HARDWARE & SOFTWARE REQUIREMENTS (5 POINTS)

Item	Specification
Host Operating System	Windows
Virtualization Software (VirtualBox/VMware)	Oracle
Ubuntu Server Version	26.04.1
VM Name	richard
RAM Allocated	4 GB
Storage Allocated	100 GB
Network Mode	NAT
SSH Client Used	openssh

3 VM CONFIGURATION

Setting	Value
VM Name	UBUNTU-SERVER
OS Type	Ubuntu (64-bit)
Memory	4000 MB
Hard Disk	100 GB
Network Adapter	Bridged

4. NETWORK CONFIGURATION

Commands Executed:

Server IP Address: 172.16.0.174

Network Interface: Physical

6. REMOTE CONNECTION

Connection Details:

Parameter	Value
Server IP	172.16.0.174
Username	richard
SSH Port	22